Experimental & Research Design Showcase

Five research design approaches — with structure, hypotheses, power analysis rationale,
and what each design can and cannot tell you.

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This document demonstrates five experimental and quasi-experimental research designs. Each exhibit includes the design logic, a worked example, power analysis considerations, threats to validity, and what the design is and is not suited to answer. Examples draw from AI adoption research, behavioral economics, and public policy evaluation.

DESIGN 01

Between-Subjects Randomized Controlled Trial

The RCT is the gold standard for causal inference. Participants are randomly assigned to treatment or control conditions, eliminating systematic confounding. The between-subjects design means each participant sees only one condition — avoiding carryover effects but requiring larger samples than within-subjects designs.

WORKED EXAMPLE

Research question: Does a peer-testimonial onboarding message increase AI platform adoption rates more than a standard feature-focused message?

Component	Specification
Hypothesis	Participants receiving peer-testimonial onboarding will show higher 7-day platform activation rates than those receiving feature-focused onboarding (one-tailed, α=.05).
Conditions	Control: standard feature walkthrough email (n=250) Treatment: peer testimonial email — same length, same CTA (n=250)
Randomization	Stratified by department and seniority level to ensure balance across conditions.
Primary outcome	Platform activation within 7 days of onboarding email (binary).
Secondary outcomes	Login frequency at 30 days; self-reported trust score (post-survey).
Analysis	Chi-square test on activation rate; logistic regression controlling for department, tenure, and prior tool use.
Power	Detecting Δ=10pp (30%→40% activation) with 80% power at α=.05 requires n=194/group. N=250/group provides buffer for attrition.
IRB / Ethics	Randomization to communication variants is standard organizational practice. No deception; no sensitive data. Exempt review likely.

WHY THIS DESIGN

Only an RCT can establish that the peer message caused the difference in activation, rather than that high-adopters self-selected into peer-facing content. Without randomization, any observed difference is confounded by pre-existing motivation.

LIMITATIONS

Cannot be used when random assignment is logistically impossible (e.g., whole teams must receive the same treatment) or ethically problematic. Requires sufficient N. External validity depends on whether the sample represents the full population of interest.

DESIGN 02

Within-Subjects / Repeated Measures

In a within-subjects design, each participant experiences all conditions. This eliminates between-person variance from the error term, dramatically increasing statistical power — the same effect is detectable with roughly half the sample of a between-subjects design. The key risk is carryover: exposure to one condition contaminates responses to subsequent conditions.

WORKED EXAMPLE

Research question: Do users rate AI-assisted document summaries differently when they know the source (AI vs. human analyst) than when they don't?

Component	Specification
Hypothesis	Participants will rate identical summaries lower when labeled 'AI-generated' than when labeled 'analyst-written,' controlling for actual quality (H■: label effect > 0).
Conditions	Each participant rates 6 summaries: · 2 labeled 'AI-generated' · 2 labeled 'written by analyst' · 2 unlabeled (baseline) Order counterbalanced via Latin square.
Stimuli	Summaries are held constant for quality across conditions — same text, different label. 3 high-quality, 3 mediocre to test label × quality interaction.
Outcome	Perceived quality rating (1–7); willingness to use in work product (binary).
Analysis	Repeated-measures ANOVA (label × quality); planned contrasts for AI vs. analyst label.
Carryover control	Minimum 2 filler summaries between each target summary; order randomized per participant.
Power	Within-subjects correlation ≈.5 assumed; Cohen's d=0.4 detectable with n=52 at 80% power.

WHY THIS DESIGN

Each participant serves as their own control — removing individual differences in general quality standards from the analysis. The label effect is cleaner because we're measuring the same person's response to the same content under different attribution conditions.

LIMITATIONS

Demand characteristics: participants may guess the hypothesis after seeing multiple conditions and adjust responses. Carryover effects require careful stimulus sequencing. Not appropriate when treatment has permanent effects (training, learning) that cannot be undone.

DESIGN 03

Survey Experiment (Vignette Factorial)

A survey experiment embeds a randomized manipulation inside a survey instrument. Respondents are randomly assigned to receive different versions of a vignette — varying one or more factors — and their responses are compared. It combines the scale of a survey with the causal logic of an experiment, making it ideal for sensitive topics where observational methods can't separate cause from effect.

WORKED EXAMPLE — 2×2 FACTORIAL DESIGN

Research question: Does the source of a policy recommendation (expert vs. peer) and its framing (gain vs. loss) independently affect building code adoption intentions?

	Gain Frame ('you could save $12K in repairs')	**Loss Frame** ('you risk $12K in damages')
Expert Source (engineer/FEMA)	Cell A n=125	Cell B n=125
Peer Source (neighbor/community member)	Cell C n=125	Cell D n=125

Component	Specification
Main effects tested	H■: Loss framing increases adoption intention vs. gain framing. H■: Expert source increases adoption intention vs. peer source.
Interaction effect	H■: Source × frame interaction — peer source may amplify loss framing through social norm activation.
Outcome	Adoption intention (1–7 Likert); willingness to invest in upgrade (binary).
Analysis	2×2 ANOVA; effect sizes (η^2); planned contrasts for interaction.
Sample	N=500 total (125/cell). ALP nationally representative panel targeting homeowners.
Power	Main effect d=0.25 detectable with 80% power at n=100/cell. 125/cell provides adequate power for interaction term.

WHY THIS DESIGN

Separates the effect of framing from the effect of source in a single study. Observational data cannot disentangle these because expert messengers tend to use evidence-based framing in practice. The factorial design provides clean estimates of each factor and their interaction at survey scale without a lab setting.

LIMITATIONS

Survey responses may not predict real behavior (intention-action gap). Vignettes simplify complex real-world decisions. Demand characteristics possible if the manipulation is transparent. Interaction effects require larger N than main effects.

DESIGN 04 Pre/Post with Control Group (Quasi-Experimental)

When randomization is not feasible, a pre/post design with a non-equivalent control group uses the difference-in-differences (DiD) framework: comparing change in the treatment group to change in the control group removes shared temporal trends (regression to the mean, external events) from the causal estimate.

WORKED EXAMPLE

Research question: Did a targeted AI literacy training program increase platform adoption rates in the departments that received it, net of organization-wide trends?

	Pre-training (T1)	Post-training (T2)	Change (T2–T1)
Treatment departments (received training)	31%	52%	+21pp
Control departments (waitlisted)	33%	38%	+5pp
DiD Estimate (causal effect of training)			**+16pp**

Component	Specification
Key assumption	Parallel trends: in the absence of training, treatment and control departments would have changed at the same rate. Validated by plotting pre-period trends.
Threats to validity	Selection bias (treatment departments may differ in motivation); spillover effects (control staff learn from treated colleagues); Hawthorne effect.
Analysis	DiD regression with department and time fixed effects; cluster-robust standard errors at department level.
Data required	Adoption rate (or login data) at two time points for all departments; training assignment records.

WHY THIS DESIGN

Organizations rarely randomize training rollouts. The DiD design extracts a credible causal estimate from naturally occurring variation in who received the program first. The control group isolates the counterfactual — what would have happened without the training.

LIMITATIONS

Parallel trends assumption is untestable for the post-period and can be violated if treatment was assigned based on anticipated need. Cannot control for unobserved department-level changes coinciding with training rollout.

DESIGN 05 Behavioral Nudge A/B Test

A nudge A/B test applies choice architecture principles — default settings, social proof, implementation intentions, loss framing — to shift behavior without restricting options. The A/B structure allows clean measurement of nudge effectiveness at scale. Rooted in Thaler & Sunstein (2008), Cialdini's social influence principles, and administrative burden theory.

WORKED EXAMPLE — THREE-ARM TEST

Research question: Which nudge type most effectively increases completion of optional AI platform training among non-adopter employees?

Arm	Nudge Type	Message Version	Mechanism	n
A (Control)	None	Standard reminder email: 'Training available.'	—	200
B	Social proof	'78% of your colleagues in Research have already completed this training.'	Descriptive norm	200
C	Implementation intention	'Block 30 minutes on your calendar this week to complete training. [Add to Calendar button]'	Planning prompt	200
D	Loss frame	'Employees who haven't completed training by [date] will miss early access to new features.'	Loss aversion	200

Component	Specification
Primary outcome	Training completion within 14 days of email (binary).
Analysis	Chi-square omnibus test across arms; pairwise comparisons with Bonferroni correction (α=.05/3=.017 per comparison).
Power	Baseline completion rate ~20%. Detecting Δ=10pp with 80% power requires n=176/arm. N=200/arm provides adequate buffer.
Ethics note	Loss framing arm requires legal/HR review — language implying penalty must be accurate and approved. Deception is not involved; all messages reflect real consequences.
Behavioral theory	Arm B targets social proof (Cialdini, 1984); Arm C targets implementation intention effect (Gollwitzer, 1999); Arm D targets loss aversion (Kahneman & Tversky, 1979).

WHY THIS DESIGN

Nudges are cheap to implement and, if effective, scale at near-zero marginal cost. The multi-arm test lets you identify which mechanism is doing the work — not just whether any intervention helps. Implementation intention theory predicts Arm C outperforms Arm B in high-intention / low-follow-through populations.

LIMITATIONS Nudge effects often attenuate over time (habituation). This test measures short-term completion — sustained behavior change requires follow-up measurement. Social proof messages require accurate figures; fabricated norms backfire.

About this document: Design examples are informed by published research and professional evaluation work at RAND Corporation, including the RAND-EDA AI adoption study, FEMA building code adoption research, and the Gates Foundation behavioral segmentation project. All examples are illustrative. Full experimental design, IRB coordination, power analysis, and statistical analysis available as project services.